

# Supplementary Document for

## Dual Balanced Large Marginal Machine for Imbalanced Data Classification

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### **This PDF file includes:**

1. Comparison of Accuracy, AUC, and F1-score of SVM, PinSVM, UPSVM, LDM, and DBUPLDM on the UCI datasets.
2. Comparison of SVM, PinSVM, UPSVM, LDM, and DBUPLDM on noise-added UCI datasets.
3. Sensitivity Analysis of Parameter  $\rho$ .
4. Proof of Theorem 2.

# 1 Comparison of Accuracy, AUC, and F1-score of SVM, PinSVM, UPSVM, LDM, and DBUPLDM on the UCI datasets.

TABLE S1.

Comparison of Accuracy, AUC, and F1-score of SVM, PinSVM, UPSVM, LDM, and DBUPLDM on the UCI datasets

| $\eta$           | Dataset     | SVM   |       |      | C     | LDM   |       |      | C     | PinSVM |       |      |      | C     | $\tau$ | UPSVM |      |       |       | C     | $\tau$ | DBUPLDM |       |       |    | C | $\tau$ |
|------------------|-------------|-------|-------|------|-------|-------|-------|------|-------|--------|-------|------|------|-------|--------|-------|------|-------|-------|-------|--------|---------|-------|-------|----|---|--------|
|                  |             | Acc   | AUC   | F1   |       | Acc   | AUC   | F1   |       | Acc    | AUC   | F1   | Acc  |       |        | AUC   | F1   | Acc   | AUC   |       |        | F1      | Acc   | AUC   | F1 |   |        |
| $\eta = 0.125$   | Monk1       | 62.73 | 61.50 | 0.63 | 0.13  | 67.59 | 66.80 | 0.69 | 0.06  | 65.28  | 64.30 | 0.66 | 0.03 | 0.20  | 66.44  | 65.10 | 0.67 | 4.00  | -0.40 | 68.29 | 72.55  | 0.75    | 1.00  | 0.70  |    |   |        |
|                  | Monk2       | 67.13 | 65.42 | 0.68 | 0.02  | 67.13 | 65.38 | 0.68 | 0.02  | 67.13  | 65.26 | 0.68 | 0.02 | -0.90 | 67.13  | 65.32 | 0.68 | 0.02  | -0.40 | 67.82 | 66.87  | 0.69    | 8.00  | -0.20 |    |   |        |
|                  | Monk3       | 81.02 | 79.18 | 0.81 | 0.06  | 84.03 | 83.45 | 0.85 | 0.13  | 84.26  | 83.32 | 0.85 | 0.02 | 1.00  | 86.34  | 85.67 | 0.87 | 0.25  | -0.40 | 88.89 | 89.21  | 0.91    | 0.50  | -0.30 |    |   |        |
|                  | Pima-Indian | 76.42 | 71.27 | 0.74 | 0.02  | 77.36 | 72.41 | 0.75 | 0.02  | 78.30  | 73.19 | 0.76 | 0.25 | -1.00 | 76.42  | 71.35 | 0.74 | 0.02  | -0.30 | 79.25 | 75.83  | 0.78    | 8.00  | 0.20  |    |   |        |
|                  | WDBC        | 86.67 | 85.23 | 0.87 | 0.06  | 85.00 | 84.17 | 0.86 | 0.06  | 86.67  | 85.31 | 0.87 | 0.06 | 0.00  | 86.67  | 85.29 | 0.87 | 0.06  | 0.00  | 87.50 | 87.66  | 0.89    | 0.06  | 0.20  |    |   |        |
|                  | Echo        | 94.04 | 92.45 | 0.93 | 2.00  | 93.38 | 91.62 | 0.92 | 8.00  | 94.04  | 92.38 | 0.93 | 2.00 | 0.00  | 94.04  | 92.42 | 0.93 | 2.00  | 0.00  | 92.72 | 90.58  | 0.91    | 4.00  | 0.70  |    |   |        |
|                  | Ionosphere  | 67.09 | 65.78 | 0.68 | 0.02  | 79.91 | 78.55 | 0.80 | 2.00  | 67.95  | 66.42 | 0.69 | 0.06 | -1.00 | 71.15  | 70.63 | 0.73 | 8.00  | -0.70 | 78.85 | 75.92  | 0.78    | 2.00  | 0.40  |    |   |        |
|                  | Heberman    | 79.29 | 78.63 | 0.80 | 0.02  | 98.82 | 98.37 | 0.99 | 0.06  | 95.86  | 95.24 | 0.96 | 0.02 | 0.10  | 95.86  | 95.31 | 0.96 | 0.02  | 0.10  | 98.82 | 98.55  | 0.99    | 0.50  | -0.20 |    |   |        |
|                  | Statlog     | 70.59 | 70.32 | 0.73 | 0.02  | 88.24 | 87.51 | 0.88 | 0.03  | 70.59  | 70.28 | 0.73 | 0.02 | 0.00  | 90.20  | 89.46 | 0.91 | 0.25  | -0.20 | 92.16 | 91.74  | 0.93    | 0.25  | -0.50 |    |   |        |
|                  | Germans     | 32.80 | 30.55 | 0.33 | 0.02  | 68.00 | 67.36 | 0.69 | 0.02  | 67.20  | 65.82 | 0.67 | 0.25 | -1.00 | 67.20  | 65.74 | 0.67 | 8.00  | -0.80 | 76.20 | 75.69  | 0.77    | 2.00  | -0.10 |    |   |        |
|                  | Australian  | 84.48 | 83.76 | 0.85 | 0.03  | 86.55 | 85.42 | 0.87 | 0.03  | 84.48  | 83.61 | 0.85 | 0.03 | 0.00  | 87.24  | 86.58 | 0.88 | 0.03  | -0.30 | 87.24 | 87.82  | 0.89    | 0.03  | -0.90 |    |   |        |
|                  | Votes       | 85.11 | 84.63 | 0.86 | 0.02  | 95.74 | 95.28 | 0.96 | 1.00  | 94.04  | 94.15 | 0.95 | 0.02 | -0.90 | 91.49  | 91.37 | 0.93 | 0.02  | -0.10 | 95.32 | 95.76  | 0.97    | 0.50  | 0.00  |    |   |        |
|                  | Diabetes    | 67.91 | 66.41 | 0.69 | 0.02  | 81.72 | 80.57 | 0.82 | 1.00  | 67.91  | 66.38 | 0.69 | 0.02 | 0.00  | 72.76  | 71.45 | 0.73 | 8.00  | -0.80 | 81.72 | 80.83  | 0.82    | 2.00  | -0.50 |    |   |        |
|                  | Fertility   | 94.00 | 93.27 | 0.94 | 0.02  | 94.00 | 93.31 | 0.94 | 0.02  | 94.00  | 93.25 | 0.94 | 0.02 | 0.00  | 94.00  | 93.30 | 0.94 | 0.02  | 0.00  | 94.00 | 94.66  | 0.95    | 0.02  | -1.00 |    |   |        |
|                  | Sonar       | 69.44 | 68.55 | 0.71 | 0.13  | 74.07 | 73.62 | 0.76 | 0.06  | 72.22  | 70.89 | 0.73 | 0.02 | -1.00 | 74.07  | 73.58 | 0.76 | 0.03  | -0.30 | 76.85 | 75.74  | 0.78    | 0.50  | 0.40  |    |   |        |
| Ecoil            | 86.61       | 85.38 | 0.87  | 0.25 | 94.49 | 94.23 | 0.95  | 0.13 | 94.49 | 94.18  | 0.95  | 0.06 | 0.90 | 94.49 | 94.21  | 0.95  | 0.25 | -0.10 | 94.49 | 94.77 | 0.96   | 0.02    | -1.00 |       |    |   |        |
| Plrx             | 67.07       | 66.42 | 0.69  | 0.02 | 67.07 | 66.35 | 0.69  | 0.02 | 67.07 | 66.28  | 0.69  | 0.02 | 0.00 | 67.07 | 66.32  | 0.69  | 0.02 | -1.00 | 70.73 | 70.59 | 0.73   | 2.00    | 0.40  |       |    |   |        |
| Ave.             | 74.73       | 71.58 | 0.75  |      | 82.59 | 78.00 | 0.76  |      | 80.15 | 76.15  | 0.76  |      |      | 81.33 | 76.30  | 0.76  |      |       | 84.25 | 82.53 | 0.84   |         |       |       |    |   |        |
| $\eta = 0.0625$  | Monk1       | 62.73 | 61.35 | 0.62 | 0.13  | 67.59 | 66.60 | 0.68 | 0.06  | 65.28  | 64.10 | 0.65 | 0.03 | 0.20  | 66.44  | 65.00 | 0.66 | 4.00  | -0.40 | 68.29 | 72.45  | 0.74    | 1.00  | 0.70  |    |   |        |
|                  | Monk2       | 67.13 | 65.32 | 0.67 | 0.02  | 67.13 | 65.28 | 0.67 | 0.02  | 67.13  | 65.16 | 0.67 | 0.02 | -0.90 | 67.13  | 65.22 | 0.67 | 0.02  | -0.40 | 67.59 | 66.77  | 0.68    | 8.00  | -0.20 |    |   |        |
|                  | Monk3       | 81.02 | 79.08 | 0.80 | 0.06  | 84.03 | 83.35 | 0.84 | 0.13  | 84.26  | 83.22 | 0.84 | 0.02 | 1.00  | 86.34  | 85.57 | 0.86 | 0.25  | -0.40 | 88.89 | 89.11  | 0.90    | 0.50  | -0.30 |    |   |        |
|                  | Pima-Indian | 76.42 | 71.17 | 0.73 | 0.02  | 77.36 | 72.31 | 0.74 | 0.02  | 78.30  | 73.09 | 0.75 | 0.25 | -1.00 | 76.42  | 71.25 | 0.73 | 0.02  | -0.30 | 79.25 | 75.73  | 0.77    | 8.00  | 0.20  |    |   |        |
|                  | WDBC        | 86.67 | 85.13 | 0.86 | 0.06  | 85.00 | 84.07 | 0.85 | 0.06  | 86.67  | 85.21 | 0.86 | 0.06 | 0.00  | 86.67  | 85.19 | 0.86 | 0.06  | 0.00  | 87.50 | 87.56  | 0.88    | 0.06  | 0.20  |    |   |        |
|                  | Echo        | 94.04 | 92.35 | 0.92 | 2.00  | 93.38 | 91.52 | 0.91 | 8.00  | 94.04  | 92.28 | 0.92 | 2.00 | 0.00  | 94.04  | 92.32 | 0.92 | 2.00  | 0.00  | 92.62 | 90.48  | 0.90    | 4.00  | 0.70  |    |   |        |
|                  | Ionosphere  | 67.09 | 65.68 | 0.67 | 0.02  | 79.81 | 78.45 | 0.79 | 2.00  | 67.95  | 66.32 | 0.68 | 0.06 | -1.00 | 71.15  | 70.53 | 0.72 | 8.00  | -0.70 | 78.75 | 75.82  | 0.77    | 2.00  | 0.40  |    |   |        |
|                  | Heberman    | 79.29 | 78.53 | 0.79 | 0.02  | 98.72 | 94.14 | 0.88 | 0.06  | 95.86  | 95.14 | 0.85 | 0.02 | 0.10  | 95.86  | 95.21 | 0.85 | 0.02  | 0.10  | 98.72 | 94.45  | 0.89    | 0.50  | -0.20 |    |   |        |
|                  | Statlog     | 70.59 | 70.22 | 0.72 | 0.02  | 88.24 | 87.41 | 0.87 | 0.03  | 70.59  | 70.18 | 0.72 | 0.02 | 0.00  | 90.20  | 89.36 | 0.90 | 0.25  | -0.20 | 92.16 | 91.64  | 0.92    | 0.25  | -0.50 |    |   |        |
|                  | Germans     | 32.80 | 30.45 | 0.32 | 0.02  | 68.00 | 67.26 | 0.68 | 0.02  | 67.20  | 65.72 | 0.66 | 0.25 | -1.00 | 67.20  | 65.64 | 0.66 | 8.00  | -0.80 | 76.20 | 75.59  | 0.76    | 2.00  | -0.10 |    |   |        |
|                  | Australian  | 84.48 | 83.66 | 0.84 | 0.03  | 86.55 | 85.32 | 0.86 | 0.03  | 84.48  | 83.51 | 0.84 | 0.03 | 0.00  | 87.14  | 86.48 | 0.87 | 0.03  | -0.30 | 87.14 | 87.72  | 0.88    | 0.03  | -0.90 |    |   |        |
|                  | Votes       | 85.11 | 84.53 | 0.85 | 0.02  | 95.64 | 95.18 | 0.85 | 1.00  | 94.04  | 94.05 | 0.84 | 0.02 | -0.90 | 91.49  | 91.27 | 0.82 | 0.02  | -0.10 | 95.22 | 95.66  | 0.89    | 0.50  | 0.00  |    |   |        |
|                  | Diabetes    | 67.91 | 66.31 | 0.68 | 0.02  | 81.62 | 80.47 | 0.81 | 1.00  | 67.91  | 66.28 | 0.68 | 0.02 | 0.00  | 72.76  | 71.35 | 0.72 | 8.00  | -0.80 | 81.62 | 80.73  | 0.81    | 2.00  | -0.50 |    |   |        |
|                  | Fertility   | 94.00 | 93.17 | 0.93 | 0.02  | 94.00 | 93.21 | 0.93 | 0.02  | 94.00  | 93.15 | 0.93 | 0.02 | 0.00  | 94.00  | 93.20 | 0.93 | 0.02  | 0.00  | 94.00 | 94.56  | 0.94    | 0.02  | -1.00 |    |   |        |
|                  | Sonar       | 69.44 | 68.45 | 0.70 | 0.13  | 74.07 | 73.52 | 0.75 | 0.06  | 72.22  | 70.79 | 0.72 | 0.02 | -1.00 | 74.07  | 73.48 | 0.75 | 0.03  | -0.30 | 76.85 | 75.64  | 0.77    | 0.50  | 0.40  |    |   |        |
| Ecoil            | 86.61       | 85.28 | 0.86  | 0.25 | 94.49 | 94.13 | 0.84  | 0.13 | 94.49 | 91.08  | 0.84  | 0.06 | 0.90 | 94.49 | 94.11  | 0.94  | 0.25 | -0.10 | 94.90 | 94.67 | 0.89   | 0.02    | -1.00 |       |    |   |        |
| Plrx             | 67.07       | 66.32 | 0.68  | 0.02 | 67.07 | 66.25 | 0.68  | 0.02 | 67.07 | 66.18  | 0.68  | 0.02 | 0.00 | 67.07 | 66.22  | 0.68  | 0.02 | -1.00 | 69.51 | 70.49 | 0.72   | 2.00    | 0.40  |       |    |   |        |
| Ave.             | 74.73       | 71.07 | 0.75  |      | 82.59 | 77.07 | 0.76  |      | 80.15 | 75.17  | 0.77  |      |      | 81.33 | 76.27  | 0.76  |      |       | 84.21 | 81.83 | 0.84   |         |       |       |    |   |        |
| $\eta = 0.03125$ | Monk1       | 62.73 | 62.21 | 0.64 | 0.13  | 67.59 | 68.05 | 0.69 | 0.06  | 65.28  | 65.30 | 0.67 | 0.03 | 0.20  | 66.44  | 66.12 | 0.70 | 4.00  | -0.40 | 68.29 | 72.38  | 0.74    | 1.00  | 0.70  |    |   |        |
|                  | Monk2       | 67.13 | 65.58 | 0.69 | 0.02  | 67.13 | 65.21 | 0.67 | 0.02  | 67.13  | 65.41 | 0.69 | 0.02 | -0.90 | 67.13  | 65.47 | 0.67 | 0.02  | -0.40 | 68.29 | 66.72  | 0.70    | 1.00  | 0.10  |    |   |        |
|                  | Monk3       | 81.02 | 79.33 | 0.80 | 0.06  | 84.03 | 83.59 | 0.84 | 0.13  | 84.26  | 83.17 | 0.86 | 0.02 | 1.00  | 86.34  | 85.82 | 0.86 | 0.25  | -0.40 | 88.89 | 89.05  | 0.92    | 0.50  | -0.30 |    |   |        |
|                  | Pima-Indian | 76.42 | 71.41 | 0.73 | 0.02  | 77.36 | 72.56 | 0.74 | 0.02  | 78.30  | 73.04 | 0.77 | 0.25 | -1.00 | 76.42  | 71.50 | 0.73 | 0.02  | -0.30 | 79.25 | 76.27  | 0.80    | 8.00  | 0.20  |    |   |        |
|                  | WDBC        | 86.67 | 85.37 | 0.86 | 0.06  | 85.00 | 84.02 | 0.87 | 0.06  | 86.67  | 85.46 | 0.86 | 0.06 | 0.00  | 86.67  | 85.14 | 0.88 | 0.06  | 0.00  | 87.50 | 87.81  | 0.88    | 0.06  | 0.20  |    |   |        |
|                  | Echo        | 94.04 | 92.31 | 0.94 | 2.00  | 93.38 | 91.77 | 0.91 | 8.00  | 94.04  | 92.52 | 0.92 | 2.00 | 0.00  | 94.04  | 92.28 | 0.94 | 2.00  | 0.00  | 92.72 | 90.72  | 0.92    | 8.00  | -0.30 |    |   |        |
|                  | Ionosphere  | 67.09 | 65.63 | 0.69 | 0.02  | 79.91 | 78.70 | 0.79 | 2.00  | 67.95  | 66.57 | 0.68 | 0.06 | -1.00 | 71.15  | 70.48 | 0.74 | 8.00  | -0.70 | 79.06 | 76.07  | 0.77    | 2.00  | 0.40  |    |   |        |
|                  | Heberman    | 79.29 | 78.77 | 0.79 | 0.02  | 98.82 | 98.22 | 1.00 | 0.06  | 95.86  | 95.39 | 0.95 | 0.02 | 0.10  | 95.86  | 95.16 | 0.97 | 0.02  | 0.10  | 98.82 | 98.40  | 0.98    | 0.50  | -0.20 |    |   |        |
|                  | Statlog     | 70.59 | 70.47 | 0.72 | 0.02  | 88.24 | 87.36 | 0.89 | 0.03  | 76.47  | 75.41 | 0.76 | 4.00 | -1.00 | 90.20  | 89.31 | 0.92 | 0.25  | -0.20 | 92.16 | 91.89  | 0.92    | 0.25  | -0.50 |    |   |        |
|                  | Germans     | 32.80 | 30.40 | 0.34 | 0.02  | 68.00 | 67.51 | 0.68 | 0.02  | 67.20  | 65.67 | 0.68 | 0.06 | -1.00 | 67.20  | 65.89 | 0.66 | 2.00  | -0.80 | 76.80 | 76.07  | 0.76    | 4.00  | 0.40  |    |   |        |
|                  | Australian  | 84.48 | 83.61 | 0.86 | 0.03  | 86.55 | 85.57 | 0.86 | 0.03  | 84.48  | 83.76 | 0.84 | 0.03 | 0.00  | 87.24  | 86.43 | 0.89 | 0.03  | -0.30 | 87.24 | 87.67  | 0.9     |       |       |    |   |        |

## 2 Comparison of SVM, PinSVM, UPSVM, LDM, and DBUPLDM on noise-added UCI datasets.

TABLE S2.

Comparison of SVM, PinSVM, UPSVM, LDM, and DBUPLDM on noise-added UCI datasets

| Dataset              | SVM         |       |       |       | LDM      |       |       |       | PinSVM   |       |       |       | UPSVM    |       |       |       | DBUPLDM  |       |       |       |      |
|----------------------|-------------|-------|-------|-------|----------|-------|-------|-------|----------|-------|-------|-------|----------|-------|-------|-------|----------|-------|-------|-------|------|
|                      | Accuracy    | AUC   | F1    | C     | Accuracy | AUC   | F1    | C     | Accuracy | AUC   | F1    | C     | Accuracy | AUC   | F1    | C     | Accuracy | AUC   | F1    | C     |      |
| $\varepsilon = 0.05$ | Monk1       | 57.87 | 59.87 | 57.37 | 0.25     | 61.11 | 63.11 | 60.61 | 4.00     | 60.65 | 62.65 | 60.15 | 0.03     | 63.43 | 65.43 | 62.93 | 4.00     | 61.34 | 63.34 | 60.84 | 2.00 |
|                      | Monk2       | 67.13 | 69.13 | 66.63 | 0.02     | 67.13 | 69.13 | 66.63 | 0.02     | 67.13 | 69.13 | 66.63 | 0.13     | 67.13 | 69.13 | 66.63 | 0.02     | 67.13 | 69.13 | 66.63 | 0.02 |
|                      | Monk3       | 72.45 | 74.45 | 71.95 | 0.50     | 72.45 | 74.45 | 71.95 | 0.50     | 73.38 | 75.38 | 72.88 | 0.25     | 73.38 | 75.38 | 72.88 | 0.25     | 71.99 | 73.99 | 71.49 | 4.00 |
|                      | Pima-Indian | 76.42 | 78.42 | 75.92 | 0.02     | 77.36 | 79.36 | 76.86 | 0.02     | 76.42 | 78.42 | 75.92 | 0.25     | 76.42 | 78.42 | 75.92 | 0.02     | 80.19 | 82.19 | 79.69 | 8.00 |
|                      | WDBC        | 84.17 | 86.17 | 83.67 | 0.06     | 85.00 | 87.00 | 84.50 | 0.03     | 84.17 | 86.17 | 83.67 | 0.06     | 84.17 | 86.17 | 83.67 | 0.06     | 85.00 | 87.00 | 84.50 | 0.13 |
|                      | Echo        | 90.73 | 92.73 | 90.23 | 0.02     | 90.73 | 92.73 | 90.23 | 0.02     | 90.73 | 92.73 | 90.23 | 0.02     | 90.73 | 92.73 | 90.23 | 0.02     | 91.39 | 93.39 | 90.89 | 8.00 |
|                      | Ionosphere  | 67.09 | 69.09 | 66.59 | 0.02     | 78.85 | 80.85 | 78.35 | 4.00     | 67.74 | 69.74 | 67.24 | 0.25     | 69.23 | 71.23 | 68.73 | 2.00     | 77.99 | 79.99 | 77.49 | 4.00 |
|                      | Heberman    | 91.72 | 93.72 | 91.22 | 0.02     | 96.45 | 98.45 | 95.95 | 8.00     | 92.90 | 94.90 | 92.40 | 0.13     | 95.86 | 97.86 | 95.36 | 0.25     | 96.45 | 98.45 | 95.95 | 0.50 |
|                      | Statlog     | 76.47 | 78.47 | 75.97 | 0.02     | 92.16 | 94.16 | 91.66 | 0.02     | 76.47 | 78.47 | 75.97 | 0.02     | 82.35 | 84.35 | 81.85 | 0.25     | 96.08 | 98.08 | 95.58 | 1.00 |
|                      | Germans     | 32.80 | 34.80 | 32.30 | 0.02     | 67.40 | 69.40 | 66.90 | 0.02     | 67.20 | 69.20 | 66.70 | 0.25     | 59.20 | 61.20 | 58.70 | 4.00     | 73.60 | 75.60 | 73.10 | 4.00 |
|                      | Australian  | 55.86 | 57.86 | 55.36 | 0.02     | 74.48 | 76.48 | 73.98 | 1.00     | 69.66 | 71.66 | 69.16 | 8.00     | 76.21 | 78.21 | 75.71 | 8.00     | 75.52 | 77.52 | 75.02 | 0.25 |
|                      | Votes       | 80.00 | 82.00 | 79.50 | 0.02     | 80.43 | 82.43 | 79.93 | 0.50     | 80.85 | 82.85 | 80.35 | 0.02     | 83.83 | 85.83 | 83.33 | 0.02     | 83.83 | 85.83 | 83.33 | 8.00 |
|                      | Diabetes    | 67.91 | 69.91 | 67.41 | 0.02     | 80.22 | 82.22 | 79.72 | 2.00     | 72.39 | 74.39 | 71.89 | 0.03     | 72.76 | 74.76 | 72.26 | 4.00     | 80.60 | 82.60 | 80.10 | 0.03 |
|                      | Fertility   | 94.00 | 96.00 | 93.50 | 0.02     | 94.00 | 96.00 | 93.50 | 0.02     | 94.00 | 96.00 | 93.50 | 0.02     | 94.00 | 96.00 | 93.50 | 0.02     | 94.00 | 96.00 | 93.50 | 0.02 |
|                      | Sonar       | 58.33 | 60.33 | 57.83 | 0.25     | 62.04 | 64.04 | 61.54 | 8.00     | 59.26 | 61.26 | 58.76 | 0.25     | 62.04 | 64.04 | 61.54 | 0.50     | 64.81 | 66.81 | 64.31 | 4.00 |
|                      | Ecoil       | 51.18 | 53.18 | 50.68 | 0.02     | 54.33 | 56.33 | 53.83 | 0.03     | 59.84 | 61.84 | 59.34 | 0.06     | 62.20 | 64.20 | 61.70 | 0.02     | 61.42 | 63.42 | 60.92 | 8.00 |
|                      | Plrx        | 67.07 | 69.07 | 66.57 | 0.02     | 67.07 | 69.07 | 66.57 | 0.02     | 67.07 | 69.07 | 66.57 | 0.02     | 68.29 | 70.29 | 67.79 | 0.06     | 70.73 | 72.73 | 70.23 | 8.00 |
|                      | Ave.        | 70.07 | 72.07 | 69.57 |          | 76.54 | 78.54 | 76.04 |          | 74.11 | 76.11 | 73.61 |          | 75.37 | 77.37 | 74.87 |          | 78.36 | 80.36 | 77.86 |      |
| $\varepsilon = 0.1$  | Monk1       | 57.64 | 59.64 | 57.14 | 0.25     | 61.11 | 63.11 | 60.61 | 4.00     | 60.42 | 62.42 | 59.92 | 0.02     | 63.43 | 65.43 | 62.93 | 4.00     | 61.34 | 63.34 | 60.84 | 2.00 |
|                      | Monk2       | 67.13 | 69.13 | 66.63 | 0.02     | 67.13 | 69.13 | 66.63 | 0.02     | 67.13 | 69.13 | 66.63 | 0.13     | 67.13 | 69.13 | 66.63 | 0.02     | 67.13 | 69.13 | 66.63 | 0.02 |
|                      | Monk3       | 71.76 | 73.76 | 71.26 | 0.25     | 72.22 | 74.22 | 71.72 | 0.50     | 73.38 | 75.38 | 72.88 | 0.50     | 73.38 | 75.38 | 72.88 | 0.50     | 71.99 | 73.99 | 71.49 | 4.00 |
|                      | Pima-Indian | 76.42 | 78.42 | 75.92 | 0.02     | 77.36 | 79.36 | 76.86 | 0.02     | 76.42 | 78.42 | 75.92 | 0.25     | 76.42 | 78.42 | 75.92 | 0.02     | 80.19 | 82.19 | 79.69 | 8.00 |
|                      | WDBC        | 84.17 | 86.17 | 83.67 | 0.06     | 85.00 | 87.00 | 84.50 | 0.03     | 84.17 | 86.17 | 83.67 | 0.06     | 84.17 | 86.17 | 83.67 | 0.06     | 85.00 | 87.00 | 84.50 | 0.13 |
|                      | Echo        | 90.73 | 92.73 | 90.23 | 0.02     | 90.73 | 92.73 | 90.23 | 0.02     | 90.73 | 92.73 | 90.23 | 0.02     | 91.39 | 93.39 | 90.89 | 0.03     | 91.39 | 93.39 | 90.89 | 8.00 |
|                      | Ionosphere  | 67.09 | 69.09 | 66.59 | 0.02     | 78.85 | 80.85 | 78.35 | 8.00     | 69.87 | 71.87 | 69.37 | 4.00     | 69.02 | 71.02 | 68.52 | 2.00     | 77.78 | 79.78 | 77.28 | 4.00 |
|                      | Heberman    | 91.72 | 93.72 | 91.22 | 0.02     | 96.45 | 98.45 | 95.95 | 8.00     | 95.27 | 97.27 | 94.77 | 0.03     | 95.86 | 97.86 | 95.36 | 0.25     | 96.45 | 98.45 | 95.95 | 0.50 |
|                      | Statlog     | 76.47 | 78.47 | 75.97 | 0.02     | 94.12 | 96.12 | 93.62 | 0.25     | 76.47 | 78.47 | 75.97 | 0.02     | 82.35 | 84.35 | 81.85 | 0.25     | 96.08 | 98.08 | 95.58 | 1.00 |
|                      | Germans     | 32.80 | 34.80 | 32.30 | 0.02     | 67.40 | 69.40 | 66.90 | 0.02     | 67.20 | 69.20 | 66.70 | 0.25     | 59.20 | 61.20 | 58.70 | 2.00     | 73.60 | 75.60 | 73.10 | 4.00 |
|                      | Australian  | 55.86 | 57.86 | 55.36 | 0.02     | 74.48 | 76.48 | 73.98 | 1.00     | 65.86 | 67.86 | 65.36 | 0.03     | 75.86 | 77.86 | 75.36 | 8.00     | 75.52 | 77.52 | 75.02 | 0.25 |
|                      | Votes       | 82.13 | 84.13 | 81.63 | 0.02     | 80.85 | 82.85 | 80.35 | 0.25     | 82.13 | 84.13 | 81.63 | 0.02     | 84.26 | 86.26 | 83.76 | 0.02     | 83.83 | 85.83 | 83.33 | 8.00 |
|                      | Diabetes    | 67.91 | 69.91 | 67.41 | 0.03     | 80.22 | 82.22 | 79.72 | 2.00     | 69.03 | 71.03 | 68.53 | 0.06     | 72.76 | 74.76 | 72.26 | 4.00     | 80.60 | 82.60 | 80.10 | 0.03 |
|                      | Fertility   | 94.00 | 96.00 | 93.50 | 0.02     | 94.00 | 96.00 | 93.50 | 0.02     | 94.00 | 96.00 | 93.50 | 0.02     | 94.00 | 96.00 | 93.50 | 0.02     | 94.00 | 96.00 | 93.50 | 0.02 |
|                      | Sonar       | 59.26 | 61.26 | 58.76 | 0.25     | 62.04 | 64.04 | 61.54 | 8.00     | 59.26 | 61.26 | 58.76 | 0.02     | 62.04 | 64.04 | 61.54 | 0.25     | 64.81 | 66.81 | 64.31 | 2.00 |
|                      | Ecoil       | 51.18 | 53.18 | 50.68 | 0.02     | 54.33 | 56.33 | 53.83 | 0.03     | 62.20 | 64.20 | 61.70 | 8.00     | 62.20 | 64.20 | 61.70 | 0.02     | 61.42 | 63.42 | 60.92 | 8.00 |
|                      | Plrx        | 67.07 | 69.07 | 66.57 | 0.02     | 67.07 | 69.07 | 66.57 | 0.02     | 67.07 | 69.07 | 66.57 | 0.02     | 68.29 | 70.29 | 67.79 | 0.06     | 70.73 | 72.73 | 70.23 | 8.00 |
|                      | Ave.        | 70.20 | 72.20 | 69.70 |          | 76.67 | 78.67 | 76.17 |          | 74.15 | 76.15 | 73.65 |          | 75.40 | 77.40 | 74.90 |          | 78.34 | 80.34 | 77.84 |      |

### 3 Sensitivity Analysis of Parameter $\rho$

This section presents the detailed experimental setup and results for the sensitivity analysis of  $\rho$ .

#### 3.1 Experimental Setup

- **Datasets:** Two datasets are selected: one is the Pharyngiti (mild imbalance, class ratio 8.26:1), and the other is the Oil Spill (severe imbalance, class ratio 21.85:1).
- **Evaluation Metrics:** Accuracy(Acc), AUC, F-measure.
- **Parameter Range:**  $\rho \in \{0, 0.2, 0.4, 0.5, 0.6, 0.7, 0.8, 1.0\}$ .

#### 3.2 Experimental Results

TABLE S3.

The performance of different balanced ratio datasets under different  $\rho$  values.

| $\rho$ | Pharyngitis (8.26:1) |             |             | Oil Spill (21.85:1) |             |             |
|--------|----------------------|-------------|-------------|---------------------|-------------|-------------|
|        | Acc                  | AUC         | F-measure   | Acc                 | AUC         | F-measure   |
| 0.0    | 0.61                 | 0.57        | 0.63        | 0.55                | 0.50        | 0.58        |
| 0.2    | 0.69                 | 0.64        | 0.71        | 0.62                | 0.56        | 0.66        |
| 0.4    | 0.78                 | 0.75        | 0.80        | 0.70                | 0.65        | 0.73        |
| 0.5    | <b>0.82</b>          | 0.79        | <b>0.84</b> | 0.76                | 0.71        | 0.79        |
| 0.6    | 0.81                 | <b>0.89</b> | 0.82        | <b>0.79</b>         | 0.74        | <b>0.83</b> |
| 0.7    | 0.77                 | 0.81        | 0.75        | 0.77                | 0.77        | 0.78        |
| 0.8    | 0.70                 | 0.72        | 0.65        | 0.68                | 0.79        | 0.61        |
| 1.0    | 0.63                 | 0.80        | 0.52        | 0.58                | <b>0.81</b> | 0.43        |

From Table S3, in the Pharyngitis dataset, the model achieves peak performance at  $\rho = 0.5$ . For the Oil Spill dataset, the optimal  $\rho$  shifts to 0.6. This aligns with the guideline in Remark 1: for low imbalance,  $\rho$  should be restricted to 0.4–0.6, while for high imbalance,  $\rho$  needs to increase to 0.6–0.7. The shift in optimal  $\rho$  demonstrates that  $\rho$  effectively modulates the balance factor’s sensitivity, allowing it to adapt to different imbalance degrees—lower  $\rho$  provides moderate intervention for mild imbalance, while higher  $\rho$  amplifies the balance factor’s effect to protect minority classes in severe imbalance.

## 4 Proof of Theorem 2

Combine Eq. (22) in the main text, the Lagrangian function of Eq. (24) in the main text is

$$\begin{aligned} L(\delta, \xi; \alpha, \beta) = & \frac{1}{2} \delta^T Q \delta + E^T \delta + \sum_{i=1}^l \rho_i \theta_i \xi_i \\ & - \sum_{i=1}^l \alpha_i (\xi_i + y_i \delta^T K_i - 1) - \sum_{i=1}^l \beta_i (\xi_i + \tau(1 - y_i \delta^T K_i)). \end{aligned} \quad (1)$$

According to the Karush–Kuhn–Tucker (KKT) optimization condition, the following equations can be obtained

$$\begin{aligned} \frac{\partial L}{\partial \delta} = & Q \delta + E - \sum_{i=1}^l \alpha_i y_i K_i + \tau \sum_{i=1}^l \beta_i y_i K_i = 0, \\ \frac{\partial L}{\partial \xi_i} = & \rho_i \theta_i - \alpha_i - \beta_i = 0, \forall i = 1, 2, \dots, l. \end{aligned} \quad (2)$$

Therefore, the dual problem of the original Eq. (24) in the main text is

$$\begin{aligned} \min_{\alpha, \beta} & \frac{1}{2} (\alpha - \tau \hat{\beta})^T A (\alpha - \tau \hat{\beta}) + \left( \frac{\lambda_1 A \theta e - l e}{l} \right)^T (\alpha - \tau \hat{\beta}) \\ \text{s.t.} & \alpha_i + \beta_i = \rho_i, \\ & \alpha_i \geq 0, \beta_i \geq 0, i = 1, \dots, l, \end{aligned} \quad (3)$$

where

$$\begin{aligned} A = & [\text{diag}(Y)] K Q^{-1} K [\text{diag}(Y)], \\ e = & [1, \dots, 1]_{1 \times m}. \end{aligned} \quad (4)$$

It can be seen from the Eq. (3) that when  $\tau$  is negative, the final optimization problem of the model is different from that when  $\tau$  is positive. This proves that Huang et al.'s method of extending  $\tau$  to negative values (?) is wrong.

We define the signum function  $v_u$  by

$$v_u = \begin{cases} 1 & \text{if } u \geq 0, \\ -1 & \text{otherwise,} \end{cases} \quad (5)$$

and let  $\hat{\beta} = |\tau| \beta$ , then the dual Eq. (3) can be given by

$$\begin{aligned} \min_{\alpha, \hat{\beta}} & \frac{1}{2} (\alpha - v_\tau \hat{\beta})^T A (\alpha - v_\tau \hat{\beta}) + \left( \frac{\lambda_1 A \theta e - l e}{l} \right)^T (\alpha - v_\tau \hat{\beta}) \\ \text{s.t.} & \alpha_i + \frac{\hat{\beta}_i}{|\tau|} = \rho_i, \\ & \alpha_i \geq 0, \hat{\beta}_i \geq 0, i = 1, \dots, l. \end{aligned} \quad (6)$$